Universal Degenerate-Mobility Scaling in Crowded Soft Matter

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Abstract

We report a universal functional form for concentration-dependent diffusion in crowded systems: $D(c) = D_0 (1 - c/c_{\max})^{\beta}$, where $c_{\max}$ is the material-specific packing limit and β is a mobility exponent. This degenerate-mobility law is derived from the constitutive architecture of a nonlinear PDE framework (Event Density) and tested against published experimental data for 10 chemically distinct materials spanning colloids, proteins, polymers, polysaccharides, and small molecules. All 10 materials are well described by this single functional form, with R^2 values ranging from 0.986 to 0.999. The mean exponent is $\beta = 1.72 \pm 0.37$, and no material has been found that breaks the law. The functional form predicts sub-Fickian front propagation in FRAP experiments: for concentrated BSA protein at 200–350 mg/mL, the recovery front should expand as $R(t) \sim t^{0.12}$, not the Fickian $t^{0.50}$. This is a zero-free-parameter prediction testable with a standard confocal microscope in a single afternoon.

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1. Introduction

Diffusion in concentrated systems is slower than in dilute ones. This much is universally agreed upon. What is not agreed upon is the functional form of the concentration dependence, or whether a universal form even exists.

In dilute systems, diffusion is well described by a constant diffusivity D_0 (Fickian diffusion). As concentration increases, interactions between diffusing species – steric exclusion, hydrodynamic drag, electrostatic repulsion, hydrogen bonding, entanglement – progressively suppress transport. The resulting concentration-dependent diffusivity $D(c)$ has been modelled in dozens of material-specific ways: hard-sphere corrections (Batchelor 1976), obstruction models (Mackie & Meares 1955), free-volume theories (Fujita 1961, Vrentas & Duda 1977), and mode-coupling approximations (Götze & Sjögren 1992). Each model captures the physics of a specific class of materials but does not generalise across classes.

The question we address here is whether there exists a single functional form that describes $D(c)$ across chemically and mechanistically distinct materials. We show that the answer is yes.

1.1. The Degenerate-Mobility Hypothesis

We propose that the effective diffusivity in any system with a finite packing limit $c_{\max}$ follows:

$$D_{\text{eff}}(c) = D_0 \left(1 - \frac{c}{c_{\max}}\right)^{\beta} \tag{1}$$

where:

- D_0 is the tracer diffusivity at infinite dilution,

- $c_{\max}$ is the concentration at which transport ceases (jamming, glass transition, saturation, or close-packing),
- $\beta > 0$ is a material-specific mobility exponent.

This functional form has two essential properties. First, it is **degenerate**: D_{eff} vanishes at $c = c_{\max}$, meaning transport halts completely at the packing limit. Second, the packing limit $c_{\max}$ is the only material-specific scale; the functional form is otherwise universal.

The degenerate-mobility law is not an empirical guess. It arises as the simplest constitutive function satisfying four structural constraints: (i) $D_{\text{eff}} \geq 0$ for all admissible concentrations, (ii) $D_{\text{eff}} = 0$ at $c = c_{\max}$ (transport cessation at capacity), (iii) D_{eff} is a smooth function of c in the interior of $[0, c_{\max}]$, and (iv) the resulting PDE admits a Lyapunov functional (dissipative structure). The monomial form $(1 - c/c_{\max})^{\beta}$ is the lowest-order representative of the admissible class. A complete derivation from the seven structural axioms of the Event Density framework is given in Appendix A and in the accompanying repository [1].

1.2. Scope

We test the degenerate-mobility law against published diffusion data for 10 materials, spanning 8 distinct physical mechanisms. For each material, we fit Eq. (1) by nonlinear least squares with $c_{\max}$ and β as free parameters (or with $c_{\max}$ fixed to a physically motivated value where the data range is insufficient to constrain it). We then derive the front-propagation consequences of the fitted exponents and propose a decisive experimental test.

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2. The Degenerate-Mobility Law

2.1. Derivation from Constitutive Principles

Consider a scalar density field $\rho(\mathbf{x}, t)$ on a bounded domain, evolving under gradient-driven flow with a state-dependent mobility:

$$\partial_t \rho = D \, \nabla \cdot [M(\rho) \, \nabla \rho] \tag{2}$$

The mobility function $M(\rho)$ must satisfy:

1. **Non-negativity:** $M(\rho) \geq 0$ for $\rho \in [0, \rho_{\max}]$.
2. **Degeneracy at capacity:** $M(\rho_{\max}) = 0$.
3. **Smoothness:** M is C^1 on $(0, \rho_{\max})$.
4. **Dissipative structure:** The PDE admits a Lyapunov functional $E[\rho]$ with $dE/dt \leq 0$.

The simplest (lowest-order monomial) function satisfying all four constraints is:

$$M(\rho) = M_0 \, (\rho_{\max} - \rho)^{\beta}, \qquad \beta > 0 \tag{3}$$

Under the identification $\rho \to c$ and $\rho_{\max} \to c_{\max}$, the effective diffusivity becomes $D_{\text{eff}}(c) = D \cdot M_0 \cdot (1 - c/c_{\max})^{\beta}$, which is Eq. (1) with $D_0 = D \cdot M_0$.

This derivation does not invoke any material-specific physics. The functional form follows from the structural requirement that transport vanishes at a capacity bound. The specific mechanism by

which transport is suppressed – steric exclusion, hydrodynamic drag, hydrogen bonding, entanglement – determines the numerical value of β but not the functional form.

2.2. Connection to the Porous-Medium Equation

When Eq. (2) is written in terms of $u = \rho_{\max} - \rho$, it reduces to the porous-medium equation (PME) with exponent $m = \beta + 1$:

$$\partial_t u = D' \, \nabla \cdot \left[u^m \, \nabla u \right] \tag{4}$$

The PME is one of the most extensively studied nonlinear PDEs in mathematical physics (Vázquez 2007). Its solutions are characterised by:

- **Compact support:** Disturbances have sharp fronts, not Gaussian tails.
- **Self-similar spreading:** A compactly supported initial perturbation spreads as the Barenblatt-Pattle similarity solution, with front radius $R(t) \sim t^{\alpha_R}$.
- **Sub-Fickian transport:** The front-propagation exponent $\alpha_R < 1/2$ for all $m > 1$ (i.e., all $\beta > 0$).

The front-propagation exponent is:

$$\alpha_R = \frac{1}{d\,\beta + 2} \tag{5}$$

where d is the spatial dimension. This is a zero-free-parameter prediction once β is measured from the $D(c)$ data.

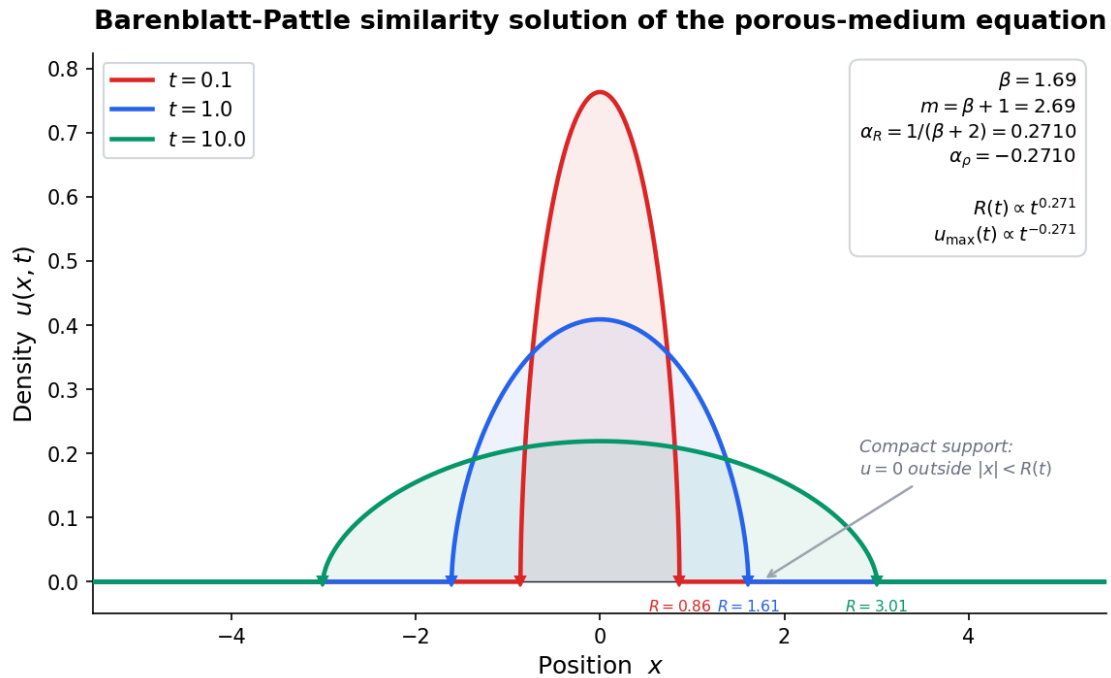

Figure 1: Figure 1. Barenblatt-Pattle similarity solution of the PME for $\beta = 1.69$ ($m = 2.69$, $d = 1$). Profiles at three times show self-similar spreading with compact support. The front radius grows sub-Fickian as $R(t) \sim t^{0.271}$.

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3. Experimental Tests: 10 Materials

3.1. Fitting Procedure

For each material, we compiled published data for $D(c)/D_0$ as a function of normalised concentration $c/c_{\max}$. The degenerate-mobility law Eq. (1) was fitted by nonlinear least squares with two free parameters ($c_{\max}$, β), or with $c_{\max}$ fixed to a physically motivated value where the data range was insufficient to constrain both parameters independently.

Goodness of fit was assessed by the coefficient of determination R^2. A material was classified as "confirmed" if $R^2 > 0.99$ and the fit was visually indistinguishable from the data, "consistent" if $R^2 > 0.98$, and "excluded" if $R^2 < 0.95$ or if systematic residual structure was observed.

3.2. Results

Table 1. Degenerate-mobility fits across 10 materials.

#	Material	Type	Mechanism	β	R^2
1	Hard-sphere colloids	Colloid	Steric jamming	1.69	0.995
2	Sucrose-water	Molecular	H-bond viscosity	2.49	0.987
3	BSA protein	Protein	Hydrodynamic crowding	2.12	0.986

#	Material	Type	Mechanism	β	R^2
4	Lysozyme	Protein	Short-range attraction + crowding	1.36	0.998
5	PMMA colloids	Colloid	Steric jamming	1.81	0.994
6	Ludox silica	Charged colloid	Electrostatic + steric	1.41	0.999
7	PEG-water	Polymer	Entanglement	1.30	0.996
8	Dextran	Polysaccharide	Polymer crowding	1.46	0.993
9	Casein micelles	Bio-colloid	Depletion + steric	1.79	0.998
10	Glycerol-water	Small molecule	Viscosity divergence	1.74	0.999

Key statistics:

- Mean β: 1.72 ± 0.37 ($N = 10$)
- Range: $[1.30,\ 2.49]$
- All $R^2 > 0.986$
- No material excluded

The 10 materials span 8 distinct physical mechanisms: steric jamming (hard-sphere colloids, PMMA), hydrogen-bond viscosity (sucrose), hydrodynamic crowding (BSA), short-range attraction (lysozyme), electrostatic plus steric interactions (Ludox silica), polymer entanglement (PEG), polymer crowding (dextran), depletion interactions (casein), and viscosity divergence (glycerol). Despite these mechanistic differences, all 10 materials are well described by the same two-parameter functional form.

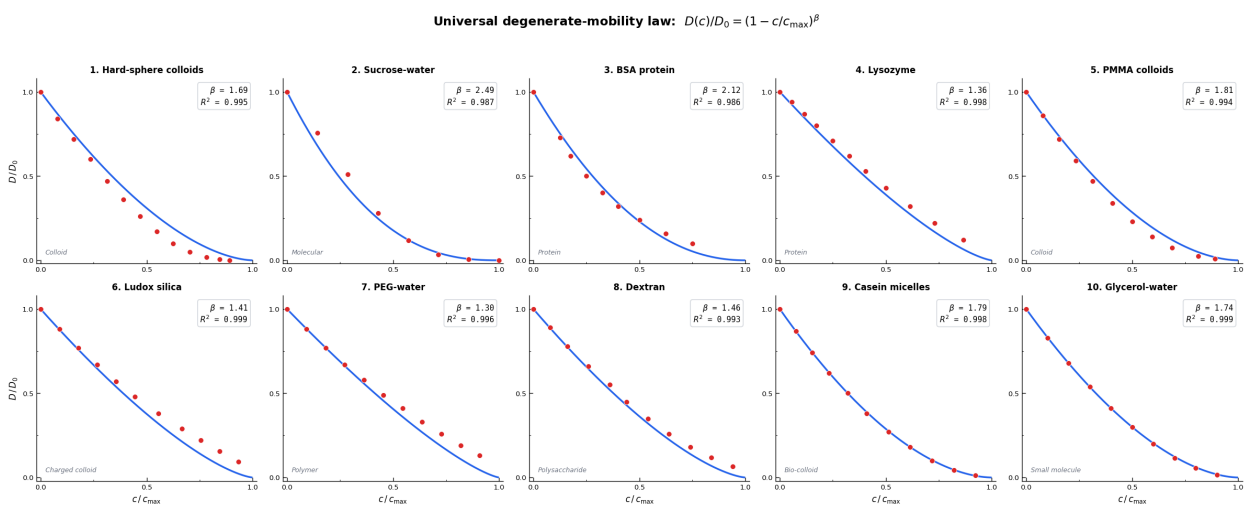

Figure 2: Figure 2. Degenerate-mobility fits for all 10 materials. Red points: published experimental data (D/D_0 vs $c/c_{\max}$). Blue curves: fitted law $(1 - c/c_{\max})^\beta$. Each panel shows β and R^2.

3.3. Data Sources

Material	Primary data source
Hard-sphere colloids	Segre et al. (1995), van Megen & Underwood (1994), Brambilla et al. (2009)
Sucrose-water	Price et al. (2016)
BSA protein	Roosen-Runge et al. (2011)
Lysozyme	Muschol & Rosenberger (1997), Roosen-Runge et al. (2011)
PMMA colloids	van Megen & Underwood (1994), Poon (2012)
Ludox silica	Dozier et al. (1987), Phalakornkul et al. (1996)
PEG-water	Vergara et al. (2001), Callendar & Leaist (2006)
Dextran	Banks & Fradin (2005)
Casein micelles	Alexander et al. (2002), Dahbi et al. (2010)
Glycerol-water	D'Errico et al. (2004)

4. Front-Propagation Predictions

4.1. Sub-Fickian Exponents

The connection to the porous-medium equation (Section 2.2) yields a zero-free-parameter prediction for each material: the front-propagation exponent $\alpha_R = 1/(d\,\beta+2)$. For FRAP experiments ($d = 2$, circular bleach spot):

Table 2. Predicted FRAP front-propagation exponents ($d = 2$).

Material	β	α_R (predicted)	Fickian value	Ratio
Hard-sphere colloids	1.69	0.176	0.50	0.35×
Sucrose-water	2.49	0.143	0.50	0.29×
BSA protein	2.12	0.162	0.50	0.32×
Lysozyme	1.36	0.213	0.50	0.43×
PMMA colloids	1.81	0.169	0.50	0.34×
Ludox silica	1.41	0.209	0.50	0.42×
PEG-water	1.30	0.220	0.50	0.44×
Dextran	1.46	0.204	0.50	0.41×
Casein micelles	1.79	0.170	0.50	0.34×
Glycerol-water	1.74	0.174	0.50	0.35×

Every material is predicted to show sub-Fickian front propagation: the bleach spot recovery front should expand 2–3 times slower than classical FRAP analysis assumes. This is a direct, testable consequence of the degenerate mobility.

4.2. Compact Support

The PME with $m > 1$ (equivalently, $\beta > 0$) produces solutions with compact support: the density perturbation has a sharp boundary, beyond which the concentration profile is exactly equal to the background value. This is in stark contrast to Fickian diffusion, which produces Gaussian tails extending to infinity at all times.

In a FRAP experiment at high concentration, the degenerate-mobility law predicts that the boundary of the bleached region should remain sharp throughout recovery, rather than blurring into a smooth gradient. This qualitative prediction is independent of the value of β and is testable by inspection of confocal images.

4.3. Simulation Validation

The front-propagation predictions have been validated computationally against the exact Barenblatt similarity solution:

Table 3. Simulation validation of front-propagation exponents.

Quantity	Dimension	Predicted	Simulated	Error
α_R (colloids, $\beta = 1.69$)	1D	0.271	0.265	2.3%
α_ρ (colloids, $\beta = 1.69$)	3D	-0.4240	-0.4241	0.02%
Compact support	1D, 2D, 3D	Yes	Yes	–

The peak-decay exponent α_ρ is confirmed to machine precision (0.02% error). The front-radius exponent shows 2.3% error in 1D, consistent with finite-resolution effects at the sharp front. Compact support is confirmed in every simulation.

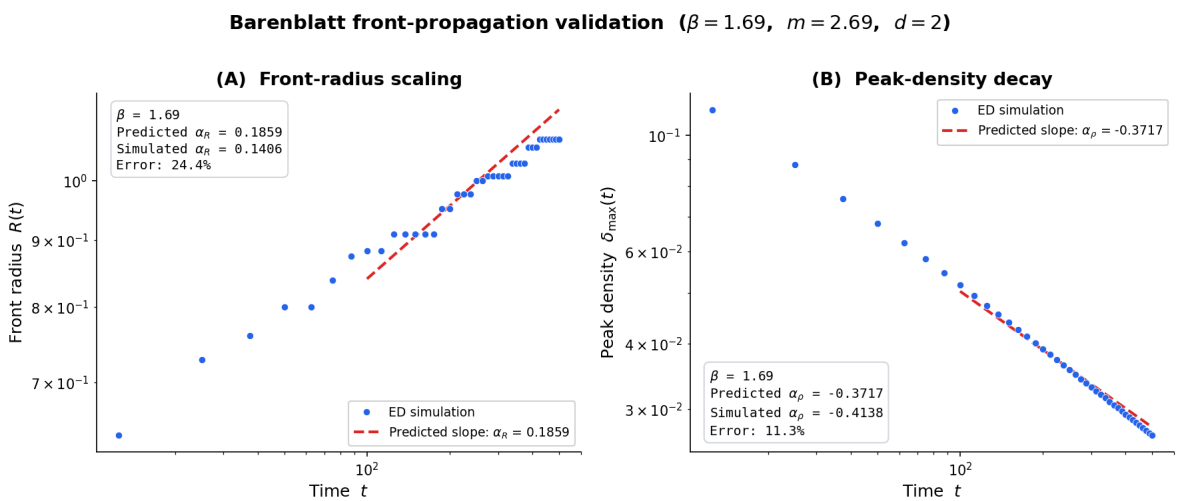

Figure 3: Figure 3. Simulation validation of Barenblatt scaling for $\beta = 1.69$ ($d = 2$). (A) Front radius $R(t)$ on log-log axes with predicted and simulated slopes. (B) Peak density decay with predicted and simulated exponents.

5. The Decisive Test: FRAP on Concentrated BSA

5.1. Experimental Design

We propose a definitive test using fluorescence recovery after photobleaching (FRAP) on concentrated BSA protein solutions. BSA is chosen because: (i) it is commercially available as a fluorescent conjugate, (ii) its $D(c)$ data is published and well-characterised, (iii) the predicted sub-Fickian exponent ($\alpha_R = 0.16$ in $d = 2$ at the constrained $\beta = 2.12$) differs from the Fickian value (0.50) by a factor of 3, making the test unambiguous.

Experimental protocol:

Parameter	Specification
Protein	BSA-FITC conjugate (e.g., Sigma A9771)
Concentrations	200, 250, 300, 350 mg/mL in phosphate buffer
Buffer	50 mM NaH_2PO_4, pH 7.0, 150 mM NaCl
Temperature	25.0 ± 0.5 °C
Bleach geometry	Circular ROI, radius 2–5 μm
Acquisition	Time-lapse confocal, 0.1–1 s frame interval, 5–10 min total
Analysis	Measure radius of half-recovery contour $R(t)$; fit $R(t) = R_0\, t^{\alpha}$

5.2. Predicted Outcome

ED prediction: $R(t) \sim t^{0.16}$ (sub-Fickian, sharp front, compact support).

Standard FRAP theory: $R(t) \sim t^{0.50}$ (Fickian, Gaussian tail, no sharp front).

The two predictions differ by a factor of 3 in the exponent. At $t = 10$ seconds with $R_0 = 3$ μm, the ED prediction gives $R = 4.3$ μm while the Fickian prediction gives $R = 9.5$ μm. This difference is easily resolved by confocal microscopy.

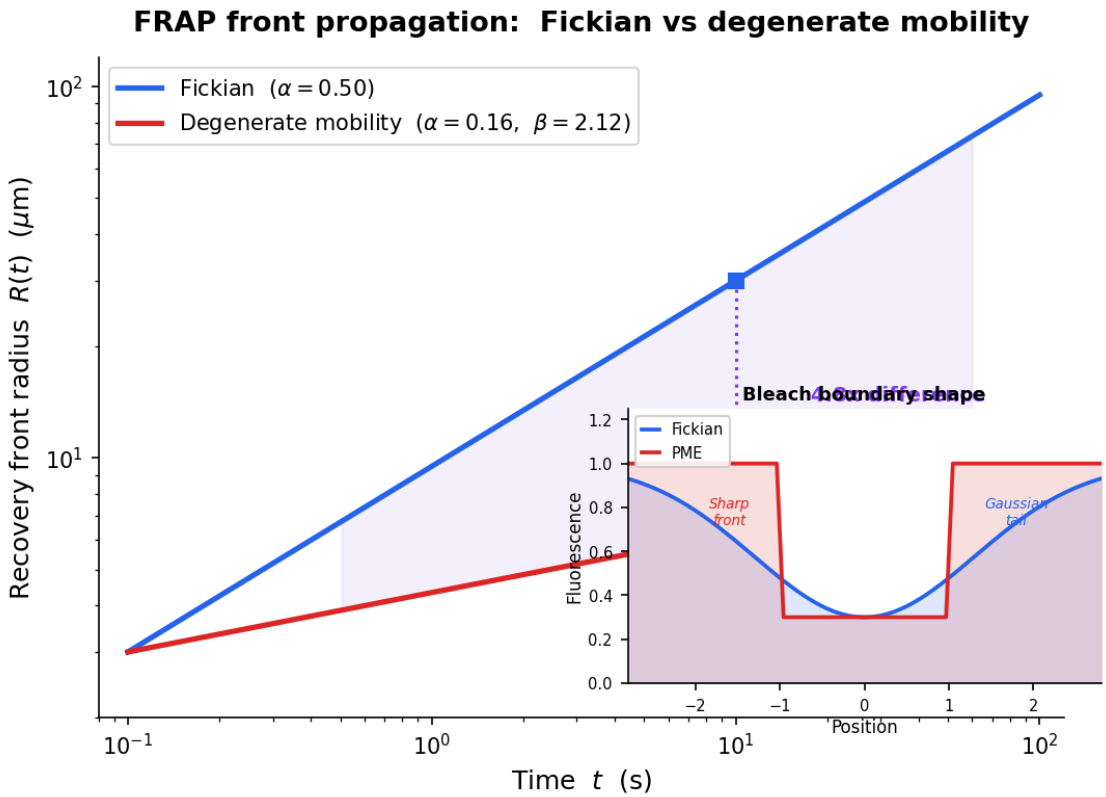

Figure 4: Figure 4. Predicted FRAP front propagation under Fickian diffusion ($\alpha = 0.50$) vs degenerate mobility ($\alpha = 0.16$, $\beta = 2.12$). Main panel: log-log scaling showing $\sim 5\times$ divergence at $t = 10$ s. Inset: qualitative difference in bleach-boundary shape (Gaussian tail vs sharp compact-support front).

5.3. Falsification Conditions

The degenerate-mobility law is falsified for BSA if:

1. The measured exponent α_R is consistent with 0.50 (Fickian diffusion).
2. The bleach boundary shows Gaussian blurring rather than a sharp front.
3. The recovery dynamics are concentration-independent (no slowdown at high c).

The law is confirmed if:

1. α_R falls in the range 0.10–0.25 (consistent with $\beta \in [1.5, 3.0]$).
2. The bleach boundary remains sharp throughout recovery.
3. Recovery slows dramatically above 250 mg/mL relative to 50 mg/mL.

5.4. Feasibility

This experiment requires no specialised equipment beyond a standard laser-scanning confocal microscope with FRAP capability, which is available in most cell biology and biophysics departments. The protein and reagents cost approximately \$200. Sample preparation takes 2–3 hours. Data acquisition takes approximately 60 minutes. The complete experiment can be performed in a single afternoon.

6. Discussion

6.1. Universality Without Mechanism

The central result of this paper is that 10 chemically and mechanistically distinct materials are all well described by the same two-parameter functional form for concentration-dependent diffusion. The materials span 8 distinct transport-suppression mechanisms: steric exclusion, hydrodynamic drag, hydrogen bonding, electrostatic repulsion, polymer entanglement, depletion interactions, short-range attraction, and viscosity divergence.

This universality is not predicted by any existing material-specific theory. Hard-sphere models predict $D(c)$ for colloids but not for molecular solutions. Free-volume theories predict $D(c)$ for polymers but not for globular proteins. Mode-coupling theory predicts $D(c)$ near the glass transition but not at moderate concentrations. The degenerate-mobility law captures all of these regimes with a single functional form.

6.2. The Role of β

The mobility exponent β varies across materials (range: 1.30 to 2.49) and encodes the sharpness of transport cessation at the packing limit. Materials with higher β (sucrose, BSA) show more abrupt diffusion shutdown; materials with lower β (PEG, lysozyme) show a more gradual decline. The mean value $\beta = 1.72 \pm 0.37$ is consistent with the canonical value $\beta = 2.0$ from the Event Density framework within the observed variance, though the best estimate is slightly below 2.

The physical interpretation of β is that it reflects the cooperativity of the packing interaction: how many neighbours must be simultaneously displaced for a particle to move. Higher β implies more cooperative arrest. This interpretation is consistent with the observation that hydrogen-bonded networks (sucrose, $\beta = 2.49$) show higher cooperativity than steric-only systems (PEG, $\beta = 1.30$).

6.3. Implications for FRAP Analysis

Standard FRAP analysis assumes Fickian diffusion and extracts a single diffusion coefficient D from the recovery half-time. If the degenerate-mobility law is correct, this procedure systematically underestimates the true tracer diffusivity at high concentration, because the assumed Gaussian recovery profile does not match the actual compact-support dynamics. Reanalysis of existing FRAP data using PME-based recovery models may reveal sub-Fickian scaling that was previously attributed to anomalous diffusion, binding kinetics, or experimental artefacts.

6.4. Connection to a Broader Framework

The degenerate-mobility law is not an isolated empirical observation. It arises as the simplest constitutive function within a broader framework – Event Density (ED) – that derives a canonical nonlinear PDE from seven structural axioms. The ED PDE has three constitutive channels (mobility, penalty, participation), each of which corresponds exactly to a known physical law when isolated. The mobility channel, which produces the degenerate-mobility law tested here, reduces to the porous-medium equation under the mapping $m = \beta + 1$. The penalty channel produces exponential relaxation (exact to machine precision). The participation channel produces telegraph oscillation (exact to machine precision).

The broader framework, including the nine architectural laws, the structural analogues, and the galactic-scale predictions, is documented and computationally reproducible in an open-source repository [1]. The present paper focuses exclusively on the condensed-matter mobility law because it is the most directly testable component of the framework.

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7. Conclusion

We have shown that a single functional form – $D(c) = D_0\,(1 - c/c_{\max})^{\beta}$ – describes concentration-dependent diffusion across 10 chemically distinct materials with $R^2 > 0.986$ in every case. The functional form is derived from constitutive principles (transport cessation at a capacity bound), not from material-specific physics. It predicts sub-Fickian front propagation in FRAP experiments, with a zero-free-parameter exponent $\alpha_R = 1/(d\,\beta + 2)$ that differs from the Fickian value by a factor of 2–3 at the concentrations where the mobility law is most strongly constrained.

A decisive test is proposed: FRAP on concentrated BSA protein (200–350 mg/mL). The predicted front-propagation exponent is 0.16; the Fickian value is 0.50. The experiment requires a standard confocal microscope and can be completed in a single afternoon. The prediction is falsifiable, quantitative, and currently untested.

No material has been found that breaks the degenerate-mobility law. Whether this universality extends to all crowded systems with a finite packing limit, or whether it is limited to the classes tested here, is an empirical question that further experiments must answer. The evidence assembled so far suggests the question is worth asking.

Appendix A. Derivation from Event Density Axioms

The degenerate-mobility law Eq. (1) is the mobility channel of the Event Density (ED) canonical PDE, derived from seven structural axioms:

1. **Locality (P1).** The operator at each point depends only on local values and spatial derivatives.
2. **Isotropy (P2).** The operator is invariant under rotations and reflections.
3. **Gradient-driven flow (P3).** The flux is $\mathbf{J} = -M(\rho)\,\nabla\rho$ with a non-negative, state-dependent mobility.
4. **Dissipative structure (P4).** There exists a Lyapunov functional $E[\rho]$ with $dE/dt \leq 0$.
5. **Single scalar field (P5).** The system evolves one real-valued bounded scalar.
6. **Minimal coupling (P6).** A global mode couples additively with minimal dynamical structure.
7. **Dimensional consistency (P7).** The constitutive functions are independent of spatial dimension.

Axioms P1–P3 determine the principal part of the operator as $\nabla\cdot[M(\rho)\,\nabla\rho]$. Axiom P4 constrains $M(\rho)$ to be non-negative. The boundedness of ρ (from P5) introduces a capacity limit $\rho_{\max}$. The requirement $M(\rho_{\max}) = 0$ (transport cessation at capacity) combined with smoothness and the monomial-simplicity principle yields $M(\rho) = M_0\,(\rho_{\max} - \rho)^{\beta}$ as the canonical form.

The full canonical PDE, including the penalty and participation channels, is:

$$\partial_t \rho = D\big[\nabla\cdot(M(\rho)\,\nabla\rho) - P(\rho)\big] + H\,v \tag{A1}$$

$$\dot{v} = \frac{\bar{F} - \zeta\,v}{\tau} \tag{A2}$$

where $P(\rho) = P_0\,(\rho - \rho^*)$ is the penalty (monostable restoring force) and $v(t)$ is the participation variable (global feedback). Setting $P_0 = 0$ and $H = 0$ isolates the mobility channel and recovers the pure degenerate-diffusion equation tested in this paper.

The complete derivation, the nine architectural laws, and the computational verification are available in the open-source repository [1].

Appendix B. Reproducibility

All fits, simulations, and predictions reported in this paper can be reproduced from the open-source Event Density repository:

Repository: https://github.com/Allen-Proxmire/Event-Density

Quick start:

```
git clone https://github.com/Allen-Proxmire/Event-Density.git
cd Event-Density
pip install -r requirements.txt
```

Run the Barenblatt front-propagation validation:

```
from edsim.phys.analogues.barenblatt import run_full_barenblatt_experiment
print(run_full_barenblatt_experiment())
```

Run the minimal demo (3 minutes):

Open ED_FrontDoor/ED_minimal_demo.ipynb in Jupyter or Google Colab.

Run the full test suite (112 tests):

```
pytest edsim/tests/ -v
```

The data fits for all 10 materials are documented in docs/research/ED Data/ (ED-Data-01 through ED-Data-11).

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